

Seminar handout

Connected and Galois Objects

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Connected and Galois Objects

Seminar notes for Talk 3

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Abstract

Let $(\mathcal{C}, F)$ be a Galois category with fibre functor $F : \mathcal{C} \rightarrow \mathbf{FSet}$. These notes discuss two basic classes of objects in $\mathcal{C}$. First, connected objects are the categorical analogue of connected finite covering spaces; every object decomposes uniquely as a finite coproduct of connected components. Second, Galois objects are the connected objects whose automorphism group acts simply transitively on the fibre. After introducing pointed objects and the domination relation, we construct the Galois closure of an arbitrary connected object. The construction is the categorical counterpart of passing from a finite separable field extension to its Galois closure, or from a connected finite covering space to its regular closure.

Standing conventions. Throughout, $(\mathcal{C}, F)$ is a Galois category in the usual sense. We use the standard axioms without restating them each time: $\mathcal{C}$ has finite limits and finite coproducts, quotients by finite group actions exist, every morphism has a strict-epimorphism–image factorization, monomorphisms identify direct summands, and F preserves the relevant finite limits, coproducts, quotients, and strict epimorphisms. We also use that F detects isomorphisms, that $F(\emptyset_{\mathcal{C}}) = \emptyset$, and that an object whose fibre has one element is final. These conventions are exactly the background results invoked below, especially in the quotient criterion for Galois objects and in the construction of the Galois closure.

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1 Connected objects

1.1 Definition

Definition 1.1 (Connected object). *A nonempty object $X \in \mathcal{C}$ is called connected if it cannot be written as a nontrivial coproduct*

$$X = X_1 \amalg X_2$$

with $X_1 \neq \emptyset_{\mathcal{C}}$ and $X_2 \neq \emptyset_{\mathcal{C}}$.

This is exactly the categorical analogue of connectedness for finite covering spaces. In the category of finite covers, a coproduct is a disjoint union, so a cover is connected precisely when it is not a disjoint union of two nonempty subcovers.

1.2 Minimality and connected components

The first result says that connected objects are the indecomposable building blocks of a Galois category; see also [1, Section 3].

Proposition 1.2 (Connected objects and connected components). *Let $X_0 \in \mathcal{C}$ be nonempty. Then X_0 is connected if and only if every monomorphism*

$$X \hookrightarrow X_0$$

with $X \neq \emptyset_{\mathcal{C}}$ is an isomorphism.

Consequently, every nonempty object $X \in \mathcal{C}$ admits a decomposition

$$X \simeq \coprod_{i=1}^r X_i$$

where each X_i is connected. This decomposition is unique up to permutation of the summands. The objects X_i are called the connected components of X .

Proof sketch. Suppose first that X_0 is not connected. Then $X_0 = X_1 \amalg X_2$ with both summands nonempty. The canonical inclusion $X_1 \hookrightarrow X_0$ is a nonempty monomorphism which is not an isomorphism.

Conversely, let $i : X \hookrightarrow X_0$ be a nonempty monomorphism. In a Galois category a monomorphism identifies its source with a direct summand of its target, so

$$X_0 \simeq X \amalg X'.$$

If X_0 is connected and X is nonempty, then X' must be empty, hence i is an isomorphism.

For the decomposition statement, if X is not connected, split it as a nontrivial coproduct. Repeating this procedure must stop because $F(X)$ is a finite set and F preserves coproducts. The uniqueness follows from the preceding minimality criterion: if

$$X \simeq \coprod_i X_i \simeq \coprod_j Y_j$$

are two decompositions into connected objects, then each inclusion $X_i \hookrightarrow X$ has nonempty image in exactly one of the components Y_j , and the minimality criterion forces $X_i \simeq Y_j$. $\square$

The finiteness of $F(X)$ is essential here. It prevents an infinite descending process of nontrivial decompositions. Applying F to a decomposition gives a disjoint decomposition of finite sets:

$$F(X) = \coprod_i F(X_i).$$

1.3 The model example: finite group sets

Let G be a profinite group and let $\mathcal{C}(G)$ be the category of finite continuous left G -sets. The fibre functor is the forgetful functor to finite sets. Then the connected objects are exactly the transitive finite G -sets. Every finite G -set decomposes uniquely as a disjoint union of orbits, and every connected object is isomorphic to

$$G/H$$

for an open subgroup $H \subseteq G$. This example should be kept in mind throughout the notes. In the next seminar talks, the main theorem will show that every pointed Galois category is equivalent to such a category of finite continuous sets for a suitable profinite group.

2 Pointed objects and domination

2.1 The pointed category

The fibre functor allows us to speak about points of objects. This is useful because maps from connected objects are determined by the image of one chosen point.

Definition 2.1 (Pointed category). *The pointed category $\mathcal{C}^{\text{pt}}$ has objects*

$$(X, \xi), \quad X \in \mathcal{C}, \quad \xi \in F(X).$$

A morphism

$$(X, \xi) \longrightarrow (Y, \eta)$$

in $\mathcal{C}^{\text{pt}}$ is a morphism $u : X \rightarrow Y$ in $\mathcal{C}$ such that

$$F(u)(\xi) = \eta.$$

Definition 2.2 (Domination). *For pointed objects (X, ξ) and (Y, η) , we say that (X, ξ) dominates (Y, η) , and write*

$$(X, \xi) \geq (Y, \eta),$$

if there exists a morphism $(X, \xi) \rightarrow (Y, \eta)$ in $\mathcal{C}^{\text{pt}}$. For unpointed objects, X dominates Y if there exists a morphism $X \rightarrow Y$ in $\mathcal{C}$.

For covering spaces, domination means that one cover lies above another cover. If $E \rightarrow S$ and $E' \rightarrow S$ are connected covers, then E dominates E' when there is a map of covers $E \rightarrow E'$.

2.2 Rigidity of maps from connected objects

Lemma 2.3 (Rigidity). *Let X_0 be connected, let $X \in \mathcal{C}$, and let $\xi_0 \in F(X_0)$. Then the evaluation map*

$$\text{ev}_{\xi_0} : \text{Hom}_{\mathcal{C}}(X_0, X) \longrightarrow F(X), \quad u \longmapsto F(u)(\xi_0)$$

is injective. Equivalently, for every $\xi \in F(X)$, there is at most one morphism

$$(X_0, \xi_0) \longrightarrow (X, \xi)$$

in $\mathcal{C}^{\text{pt}}$.

Proof. Let $u, v : X_0 \rightarrow X$ satisfy

$$F(u)(\xi_0) = F(v)(\xi_0).$$

Let $E \hookrightarrow X_0$ be the equalizer of u and v . Since F preserves finite limits, $F(E)$ is the equalizer of $F(u)$ and $F(v)$. In particular, $\xi_0 \in F(E)$, so E is nonempty. Because X_0 is connected, the monomorphism $E \hookrightarrow X_0$ is an isomorphism by Proposition 1.2. Hence $u = v$. $\square$

This is the categorical form of the uniqueness of lifts in covering space theory: a map from a connected cover is determined by the image of one point in one fibre.

2.3 Common connected refinements

The following domination result is one of the basic directedness properties of pointed Galois categories; see [1, Section 3].

Proposition 2.4 (Domination by connected pointed objects). *Let*

$$(X_i, \xi_i) \in \mathcal{C}^{\text{pt}}, \quad i = 1, \dots, r,$$

be finitely many pointed objects. Then there exists a connected pointed object (X_0, ξ_0) such that

$$(X_0, \xi_0) \geq (X_i, \xi_i)$$

for all i .

In particular, for every object $X \in \mathcal{C}$ there exists a connected pointed object (X_0, ξ_0) such that

$$\text{ev}_{\xi_0} : \text{Hom}_{\mathcal{C}}(X_0, X) \longrightarrow F(X)$$

is bijective.

Proof. For the first assertion, form the product

$$P = X_1 \times \cdots \times X_r.$$

The tuple $(\xi_1, \dots, \xi_r)$ is a point of

$$F(P) = F(X_1) \times \cdots \times F(X_r).$$

Decompose P into connected components. Since F preserves finite coproducts, the point $(\xi_1, \dots, \xi_r)$ lies in the fibre of a unique connected summand. Let X_0 be this connected component and let $\xi_0 \in F(X_0)$ be the corresponding point. The projections $P \rightarrow X_i$ restrict to morphisms

$$(X_0, \xi_0) \longrightarrow (X_i, \xi_i).$$

Thus (X_0, ξ_0) dominates all the given pointed objects.

For the second assertion, first suppose $F(X) \neq \emptyset$. Apply the first part to the finite family of pointed objects

$$(X, \xi), \quad \xi \in F(X).$$

The resulting connected pointed object (X_0, ξ_0) admits a pointed morphism to each (X, ξ) , so the evaluation map is surjective. It is injective by the rigidity lemma. Hence it is bijective.

If $F(X) = \emptyset$, then $X \simeq \emptyset_{\mathcal{C}}$. Take any connected pointed object, for instance a final object with its unique point. There is no morphism from a nonempty object to $\emptyset_{\mathcal{C}}$, since applying F would give a map from a nonempty finite set to the empty set. Hence both $\text{Hom}_{\mathcal{C}}(X_0, X)$ and $F(X)$ are empty, and the evaluation map is again bijective. $\square$

2.4 Morphisms involving connected objects

Proposition 2.5 (Basic consequences for connected objects). *The following statements hold.*

- (1) *If X_0 is connected and $u : X \rightarrow X_0$ is a morphism with $X \not\cong \emptyset_{\mathcal{C}}$, then u is a strict epimorphism.*
- (2) *If $u : X_0 \rightarrow X$ is a strict epimorphism and X_0 is connected, then X is connected.*
- (3) *If X_0 is connected, then every endomorphism $u : X_0 \rightarrow X_0$ is an automorphism.*

Proof. For (1), factor u as

$$X \twoheadrightarrow Y \hookrightarrow X_0 = Y \amalg Y'.$$

Since X is nonempty, the image Y is nonempty. As X_0 is connected, the complementary summand Y' must be empty. Therefore $Y \simeq X_0$, and u is a strict epimorphism.

For (2), suppose $X = X_1 \amalg X_2$ with both X_i nonempty. Pulling this decomposition back along u gives

$$X_0 \simeq (X_0 \times_X X_1) \amalg (X_0 \times_X X_2).$$

Since u is a strict epimorphism, $F(u)$ is surjective, so both pullbacks have nonempty fibre. This contradicts connectedness of X_0 . Hence X is connected.

For (3), an endomorphism $u : X_0 \rightarrow X_0$ is a strict epimorphism by (1). Hence $F(u)$ is a surjective map from the finite set $F(X_0)$ to itself, and therefore is bijective. Since F detects isomorphisms, u is an automorphism. $\square$

3 Galois objects

3.1 Definition

Let X_0 be connected and choose a point $\xi_0 \in F(X_0)$. The automorphism group $\text{Aut}_{\mathcal{C}}(X_0)$ acts on the finite set $F(X_0)$. By the rigidity lemma, the evaluation map

$$\text{ev}_{\xi_0} : \text{Aut}_{\mathcal{C}}(X_0) \longrightarrow F(X_0), \quad \omega \longmapsto F(\omega)(\xi_0)$$

is injective. Thus the action is always free. A Galois object is one for which this free action is also transitive.

Definition 3.1 (Galois object). *A connected object $X_0 \in \mathcal{C}$ is called Galois if for every $\xi_0 \in F(X_0)$ the map*

$$\text{ev}_{\xi_0} : \text{Aut}_{\mathcal{C}}(X_0) \longrightarrow F(X_0)$$

is bijective.

Equivalently, X_0 is Galois if $\text{Aut}_{\mathcal{C}}(X_0)$ acts simply transitively on the fibre $F(X_0)$. Since the action is free, surjectivity for one point is already equivalent to surjectivity for every point; this is made explicit in the next proposition. In the topological model, this says that the deck transformation group of a connected cover acts simply transitively on each fibre. Such covers are also called regular or normal covers.

3.2 Equivalent characterizations

Proposition 3.2 (Equivalent forms of the Galois condition). *Let X_0 be connected. The following conditions are equivalent.*

- (1) X_0 is Galois.
- (2) For some, equivalently every, $\xi_0 \in F(X_0)$, the group $\text{Aut}_{\mathcal{C}}(X_0)$ acts transitively on $F(X_0)$.
- (3) For some, equivalently every, $\xi_0 \in F(X_0)$, the group $\text{Aut}_{\mathcal{C}}(X_0)$ acts simply transitively on $F(X_0)$.
- (4) $|\text{Aut}_{\mathcal{C}}(X_0)| = |F(X_0)|$.
- (5) The quotient $X_0/\text{Aut}_{\mathcal{C}}(X_0)$ is a final object of $\mathcal{C}$.

Proof. The evaluation map $\text{Aut}_{\mathcal{C}}(X_0) \rightarrow F(X_0)$ is injective by rigidity, for every choice of ξ_0 . Hence the action of $\text{Aut}_{\mathcal{C}}(X_0)$ on $F(X_0)$ is free. For a free action of a finite group on a finite set, transitivity, simple transitivity, and equality of cardinalities are equivalent. This proves the equivalence of (1)–(4).

For (5), use the compatibility of F with quotients by finite group actions:

$$F(X_0/\text{Aut}_{\mathcal{C}}(X_0)) \simeq F(X_0)/\text{Aut}_{\mathcal{C}}(X_0).$$

The quotient set has one element exactly when the action is transitive. Since an object of a Galois category is final precisely when its fibre has one element, this is equivalent to (2). $\square$

Remark. The quotient criterion is useful because it shows that the Galois condition is intrinsic within the Galois category structure. Although the definition was phrased in terms of the fibre functor, the condition can be expressed by saying that the quotient by the automorphism group is final.

3.3 Examples

Example 3.3 (Finite covering spaces). *Let S be a path-connected, locally path-connected, and semilocally simply connected base space. Let $E \rightarrow S$ be a connected finite covering space. Then E is Galois in the above categorical sense if and only if the group of deck transformations of E acts transitively on one, hence every, fibre. This is the usual notion of a regular finite cover.*

Example 3.4 (Finite continuous G -sets). *Let G be a profinite group and let $\mathcal{C}(G)$ be the category of finite continuous left G -sets. A connected object has the form G/H for an open subgroup $H \subseteq G$. A G -equivariant automorphism of G/H is determined by the image of the coset H . With the convention*

$$\varphi_n(gH) = gn^{-1}H \quad (n \in N_G(H)),$$

composition gives a genuine group isomorphism

$$N_G(H)/H \xrightarrow{\sim} \text{Aut}_G(G/H), \quad nH \mapsto \varphi_n.$$

This group acts transitively on G/H if and only if $N_G(H) = G$, equivalently if and only if H is normal in G . Thus the Galois objects in $\mathcal{C}(G)$ are precisely the finite quotients

$$G/H$$

with $H \subseteq G$ open and normal.

4 The Galois closure

The central result of these notes is the existence of a Galois closure for every connected object. This result is the categorical analogue of taking the normal closure of a finite field extension or the regular closure of a connected covering space; see [1, Section 3].

4.1 Statement

Proposition 4.1 (Galois closure). *Let $X \in \mathcal{C}$ be connected. Then there exists a Galois object $\widehat{X}$ together with a morphism*

$$p : \widehat{X} \longrightarrow X$$

with the following minimality property: if Y is a Galois object and $q : Y \rightarrow X$ is any morphism, then there exists a morphism

$$h : Y \longrightarrow \widehat{X}$$

compatible with the maps to X , that is, $p \circ h = q$. The object $\widehat{X}$ is unique up to isomorphism and is called the Galois closure of X .

The phrase “minimal” is meant with respect to domination. Thus $\widehat{X}$ is not minimal by cardinality among all objects of $\mathcal{C}$; rather, every Galois object lying above X also lies above $\widehat{X}$.

4.2 Construction

Let X be connected. By Proposition 2.4, choose a connected pointed object (X_0, ξ_0) such that the evaluation map

$$\mathrm{ev}_{\xi_0} : \mathrm{Hom}_{\mathcal{C}}(X_0, X) \longrightarrow F(X)$$

is bijective. Write

$$\mathrm{Hom}_{\mathcal{C}}(X_0, X) = \{u_1, \dots, u_n\}.$$

Then the points

$$\xi_i := F(u_i)(\xi_0), \quad i = 1, \dots, n,$$

are exactly the elements of $F(X)$.

Consider the product X^n and the morphism

$$\pi = (u_1, \dots, u_n) : X_0 \longrightarrow X^n.$$

Apply the image factorization axiom to π :

$$X_0 \xrightarrow{\pi'} \widehat{X} \xrightarrow{\pi''} X^n,$$

where π' is a strict epimorphism and π'' identifies $\widehat{X}$ with a direct summand of X^n . We define $\widehat{X}$ to be this image object.

Let

$$\widehat{\xi}_0 := F(\pi')(\xi_0) \in F(\widehat{X}).$$

Under the inclusion $F(\widehat{X}) \subseteq F(X^n) = F(X)^n$, this point is

$$\widehat{\xi}_0 = (\xi_1, \dots, \xi_n).$$

The object $\widehat{X}$ is connected because it is the strict epimorphic image of the connected object X_0 . Moreover, $\widehat{X}$ dominates X . From now on we fix the map

$$p := \mathrm{pr}_1 \circ \pi'' : \widehat{X} \longrightarrow X^n \longrightarrow X.$$

Idea of the construction. The map $\pi : X_0 \rightarrow X^n$ records all morphisms from X_0 to X at the same time. The image $\widehat{X} \subseteq X^n$ is the smallest part of the product through which this simultaneous record factors. This construction forces enough symmetry to make $\widehat{X}$ a Galois object.

4.3 Proof that the closure is Galois

Since $\widehat{X}$ is connected, the evaluation map

$$\mathrm{ev}_{\widehat{\xi}_0} : \mathrm{Aut}_{\mathcal{C}}(\widehat{X}) \longrightarrow F(\widehat{X})$$

is injective. It remains to prove that it is surjective.

Let $\zeta \in F(\widehat{X})$ be arbitrary. Since $\pi' : X_0 \rightarrow \widehat{X}$ is a strict epimorphism, the map $F(\pi')$ is surjective. Choose $\theta \in F(X_0)$ such that

$$F(\pi')(\theta) = \zeta.$$

By Proposition 2.4, there exists a connected pointed object (Y, η) dominating both (X_0, ξ_0) and (X_0, θ) . Thus we have morphisms

$$a : (Y, \eta) \longrightarrow (X_0, \xi_0), \quad b : (Y, \eta) \longrightarrow (X_0, \theta).$$

Since Y is nonempty and X_0 is connected, both a and b are strict epimorphisms by Proposition 2.5(1).

For each i , the composite $u_i \circ b : Y \rightarrow X$ sends η to the point

$$F(u_i)(\theta) \in F(X).$$

Because the evaluation map $\mathrm{Hom}_{\mathcal{C}}(X_0, X) \rightarrow F(X)$ at ξ_0 is bijective, there is a unique index $\sigma(i)$ such that

$$F(u_{\sigma(i)})(\xi_0) = F(u_i)(\theta).$$

The two morphisms $u_i \circ b$ and $u_{\sigma(i)} \circ a$ from the connected object Y to X have the same value at η . By rigidity,

$$u_i \circ b = u_{\sigma(i)} \circ a.$$

The indices $\sigma(i)$ are all distinct. Indeed, if $\sigma(i) = \sigma(j)$, then $F(u_i)(\theta) = F(u_j)(\theta)$. Applying the rigidity lemma with the base point $\theta \in F(X_0)$ gives $u_i = u_j$, and hence $i = j$. Thus σ is a permutation of $\{1, \dots, n\}$.

Let $s_\sigma : X^n \rightarrow X^n$ denote the automorphism that permutes the factors according to σ , with the convention that

$$\mathrm{pr}_i \circ s_\sigma = \mathrm{pr}_{\sigma(i)}.$$

The equalities above say exactly that

$$\pi \circ b = s_\sigma \circ \pi \circ a.$$

Since a and b are strict epimorphisms, precomposing with either one does not change the image subobject in X^n . Hence the images of $\pi \circ a$ and $\pi \circ b$ are both the image of π , namely $\widehat{X}$. By uniqueness of image factorization, the automorphism s_σ sends the direct summand $\widehat{X} \subseteq X^n$ to itself. Hence it restricts to an automorphism

$$\omega : \widehat{X} \xrightarrow{\sim} \widehat{X}.$$

Evaluating $\pi \circ b = s_\sigma \circ \pi \circ a$ at η gives

$$F(s_\sigma)(\xi_1, \dots, \xi_n) = F(\pi)(\theta).$$

Under the inclusion $F(\widehat{X}) \subseteq F(X^n)$, the right hand side is exactly $F(\pi')(\theta) = \zeta$. Therefore the restricted automorphism ω satisfies

$$F(\omega)(\widehat{\xi}_0) = \zeta.$$

Thus every point of $F(\widehat{X})$ is in the $\text{Aut}_{\mathcal{C}}(\widehat{X})$ -orbit of $\widehat{\xi}_0$. The evaluation map is surjective, so $\widehat{X}$ is Galois.

Remark. The proof has the same flavour as the construction of a normal closure in group theory. A point ζ of the fibre of $\widehat{X}$ reorders the maps $u_i : X_0 \rightarrow X$. This reordering produces a permutation of the factors of X^n , and the image construction ensures that the permutation preserves $\widehat{X}$.

4.4 Proof of minimality

Let Y be a Galois object and let

$$q : Y \longrightarrow X$$

be a morphism. We prove that Y dominates $\widehat{X}$ over X .

Choose a point $\eta_1 \in F(Y)$ satisfying

$$F(q)(\eta_1) = \xi_1.$$

Such a point exists because q is a morphism from a nonempty object to the connected object X , hence q is a strict epimorphism by Proposition 2.5(1) and $F(q)$ is surjective. For each i , choose $\eta_i \in F(Y)$ such that

$$F(q)(\eta_i) = \xi_i.$$

For $i = 1$ we keep the chosen point η_1 and take $\gamma_1 = \text{id}_Y$. Since Y is Galois, $\text{Aut}_{\mathcal{C}}(Y)$ acts transitively on $F(Y)$; therefore for each i there is an automorphism $\gamma_i \in \text{Aut}_{\mathcal{C}}(Y)$ such that

$$F(\gamma_i)(\eta_1) = \eta_i.$$

Define

$$\kappa = (q \circ \gamma_1, \dots, q \circ \gamma_n) : Y \longrightarrow X^n.$$

Then

$$F(\kappa)(\eta_1) = (\xi_1, \dots, \xi_n) = \widehat{\xi}_0.$$

Let

$$Y \twoheadrightarrow Z \hookrightarrow X^n$$

be the image factorization of κ . Since Y is connected and $Y \twoheadrightarrow Z$ is a strict epimorphism, the object Z is connected. Moreover, the fibre of Z contains the point $\widehat{\xi}_0$.

The object $\widehat{X}$ is also a connected direct summand of X^n whose fibre contains $\widehat{\xi}_0$. By uniqueness of the connected-component decomposition of X^n , we have $Z \simeq \widehat{X}$ as direct summands of X^n . Therefore κ factors through $\widehat{X}$:

$$h : Y \longrightarrow \widehat{X} \hookrightarrow X^n.$$

Since $\gamma_1 = \text{id}_Y$ and $p = \text{pr}_1 \circ \pi''$, the first projection gives

$$p \circ h = \text{pr}_1 \circ \kappa = q \circ \gamma_1 = q.$$

Thus h is compatible with the given maps to X . This gives the desired domination $Y \rightarrow \widehat{X}$ over X , and hence $\widehat{X}$ is minimal among Galois objects dominating X .

4.5 Uniqueness

Suppose $\widehat{X}$ and $\widehat{X}'$ are two Galois closures of X . By minimality, there are morphisms

$$\widehat{X} \longrightarrow \widehat{X}', \quad \widehat{X}' \longrightarrow \widehat{X}$$

compatible with the maps to X . Their composites are endomorphisms of connected objects. By Proposition 2.5(3), these endomorphisms are automorphisms. Hence $\widehat{X}$ and $\widehat{X}'$ are isomorphic. This proves that the Galois closure is unique up to isomorphism, though not canonically unique in general.

5 Interpretations and examples of the Galois closure

5.1 Finite continuous G -sets

Let G be a profinite group and let $X = G/H$ be a connected finite continuous G -set, where $H \subseteq G$ is open. The Galois closure of X is

$$\widehat{X} = G / \text{Core}_G(H), \quad \text{Core}_G(H) = \bigcap_{g \in G} gHg^{-1}.$$

Because H is open, the action of G on the finite set G/H has finite image, so $\text{Core}_G(H)$ is an open normal subgroup of G . The natural quotient map

$$G / \text{Core}_G(H) \longrightarrow G/H$$

shows that $\widehat{X}$ dominates X . It is Galois because $\text{Core}_G(H)$ is normal. It is minimal because every open normal subgroup of G contained in H is contained in $\text{Core}_G(H)$.

This example explains why the categorical construction used all maps $X_0 \rightarrow X$ at once. In the group-set model, those maps encode the conjugates of the stabilizer, and the image construction corresponds to intersecting all conjugates.

5.2 Finite covering spaces

Let $p : E \rightarrow S$ be a connected finite covering of a path-connected, locally path-connected, and semilocally simply connected base space S . Choose a base point $s \in S$ and a point $e \in E_s$. The cover corresponds to a finite-index subgroup

$$H \subseteq \pi_1(S, s).$$

The cover is Galois precisely when H is normal in $\pi_1(S, s)$. The Galois closure corresponds to the subgroup

$$\text{Core}_{\pi_1(S, s)}(H) = \bigcap_{\gamma \in \pi_1(S, s)} \gamma H \gamma^{-1}.$$

This subgroup is the largest normal subgroup contained in H . The associated finite cover is a regular cover dominating E , and it is minimal with this property.

5.3 Analogy with field extensions

The same picture is visible in classical Galois theory. A finite separable field extension L/K need not be Galois. Its Galois closure is obtained by adjoining all conjugates of a primitive element, or equivalently by passing to the smallest normal extension of K containing L . In the categorical construction, the product X^n plays the role of collecting all conjugates at once, and the image $\widehat{X}$ is the finite object cut out by this simultaneous collection.

References

- [1] Anna Cadoret, *Galois Categories*, in *Arithmetic and Geometry Around Galois Theory*, Birkhäuser, 2013, pp. 171–246.
- [2] Alexander Grothendieck and Michèle Raynaud, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics 224, Springer, 1971.
- [3] Tamás Szamuely, *Galois Groups and Fundamental Groups*, Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2009.